

Data 510
Capstone project proposal

***Forecasting U.S. Unemployment Using
Macroeconomic Indicators: A Data
Engineering, Statistical Analysis, and Machine
Learning Project***

Project Stakeholders

Owner Team: Manish R. Kallu

Owner Product Lead: Manish R. Kallu

Peer Stakeholder Product Owners:

- Brandon Smith
- Jon Garrow
- Jackson Garro

Instructor / Sponsor: Lucas P. Cordova, Ph.D.

GitHub Repository

- <https://github.com/manishkallu01-wq/DATA510-Session-2/tree/main>

GitHub Project Board

- <https://github.com/users/manishkallu01-wq/projects/1>

Discord Category

- <https://discord.com/channels/1277725100816203942/1508588739063054376>

Studio Session: 2

Studio Formation Date: May 25, 2026

1. Problem Framing and Research Motivation

1.1 Background

The unemployment rate is one of the most important indicators used to evaluate the health of the United States economy. Government agencies, businesses, investors, workforce development organizations, researchers, and job seekers rely on labor market information to understand economic conditions and support decision-making. Changes in unemployment influence hiring activity, workforce planning, consumer confidence, business investment, and economic policy decisions.

The original concept for this project focused on analyzing historical unemployment trends in the United States. During the charter and proposal development phases, publicly available datasets and academic literature were reviewed to evaluate whether a more consequential and analytically rigorous research problem could be addressed using available economic data.

This investigation revealed that unemployment is influenced by multiple macroeconomic forces including inflation, monetary policy, economic growth, consumer sentiment, and recessionary conditions. Prior research suggested that these indicators often provide useful information regarding future labor market outcomes and may improve forecasting performance when incorporated into predictive models.

Based on findings from the literature review and the availability of high-quality public datasets, the project scope was refined from a descriptive unemployment study into a predictive and explanatory investigation of labor market conditions using integrated macroeconomic indicators.

This revised direction better utilizes the available data, aligns with the objectives of the DATA 510 Capstone, and provides an opportunity to develop a comprehensive capstone project based entirely on real-world economic data spanning nearly seventy years of United States economic history.

1.2 Why This Problem Matters

Labor market conditions affect millions of individuals, businesses, and institutions across the United States. Employers use labor market information to make hiring decisions, governments use it to evaluate economic policy, and workforce organizations use it to allocate training and development resources.

Although unemployment is widely monitored, it is generally considered a lagging indicator because labor market deterioration often becomes visible only after broader economic conditions have already changed.

At the same time, economic indicators such as inflation, interest rates, GDP growth, consumer sentiment, and recession measures are continuously monitored by government agencies and research institutions. Understanding whether these indicators provide useful information regarding future unemployment conditions may improve economic awareness, workforce planning, and labor market forecasting.

The project therefore addresses a consequential problem with direct relevance to policymakers, employers, researchers, and workforce development organizations.

1.3 Literature Review

To evaluate the feasibility and relevance of the proposed research problem, an initial review of academic literature related to labor market forecasting, economic indicators, consumer sentiment, and recession analysis was conducted.

Barnichon (2013) demonstrated that labor market dynamics can improve unemployment forecasting performance when compared with models relying exclusively on historical unemployment observations. The study highlighted the value of incorporating broader labor market information into forecasting frameworks.

Howrey (2001) found that consumer sentiment contains useful information regarding future economic activity and economic expectations. The research suggested that changes in consumer confidence may provide early signals regarding future economic conditions.

Chung, Fallick, Nekarda, and Ratner (2014) emphasized the importance of evaluating labor market conditions using multiple indicators rather than relying solely on unemployment statistics. Their

findings demonstrated that labor market behavior reflects a complex interaction of economic factors.

Favara, Gilchrist, Lewis, and Zakrajšek (2022) demonstrated that financial and macroeconomic indicators contain valuable information regarding future economic conditions and recession risk. Their work reinforced the importance of integrating multiple economic signals when evaluating future outcomes.

Collectively, these studies suggest that unemployment is influenced by multiple dimensions of economic activity and that forecasting performance may improve when broader macroeconomic indicators are incorporated into analytical frameworks. This observation motivates the current project, which integrates multiple economic indicators within a single analytical framework to evaluate both their historical relationships with unemployment and their forecasting value.

1.4 Research Gap

Existing studies have explored unemployment forecasting, recession prediction, consumer sentiment, labor market dynamics, and macroeconomic forecasting independently. However, many studies focus on a limited set of indicators, specific economic periods, or narrowly defined forecasting objectives.

Additionally, much of the existing literature emphasizes predictive performance while providing limited discussion regarding reproducible Data Engineering workflows, feature engineering strategies, visualization, stakeholder communication, and end-to-end analytical pipelines.

This project addresses these limitations by constructing a unified analytical framework that integrates six publicly available macroeconomic datasets spanning nearly seventy years of U.S. economic history. The resulting framework combines Data Engineering, statistical analysis, machine learning, visualization, and communication into a single reproducible workflow.

1.5 Consequential Problem Statement

Economic stakeholders frequently rely on unemployment statistics to evaluate labor market conditions. Because unemployment often reacts after broader economic changes have already occurred, stakeholders may have limited visibility into future labor market conditions.

Meanwhile, a substantial amount of publicly available macroeconomic information is continuously produced by organizations such as the Bureau of Labor Statistics, Federal Reserve, Bureau of Economic Analysis, University of Michigan, and National Bureau of Economic Research.

The challenge is determining whether these indicators provide meaningful information regarding future unemployment conditions and whether they can improve labor market forecasting performance.

Answering this question has implications for workforce planning, economic forecasting, business strategy, labor market analysis, and public policy decision-making.

1.6 Research Questions

Primary Research Question

To what extent can publicly available macroeconomic indicators improve forecasting of future U.S. unemployment across 3-month, 6-month, and 12-month forecasting horizons?

This question focuses on predictive analytics and machine learning. The objective is to evaluate whether macroeconomic indicators improve forecasting performance relative to historical unemployment trends alone.

Secondary Research Question

Which macroeconomic indicators demonstrate the strongest and most consistent historical relationships with unemployment across U.S. economic cycles between 1956 and 2025?

This question focuses on statistical analysis and visualization. The objective is to identify which indicators historically move most closely with unemployment and how those relationships vary across different economic environments.

1.7 Expected Outcomes

The project is expected to produce:

- A validated and reproducible macroeconomic analytical dataset.
- Statistical evidence describing relationships between economic indicators and unemployment.
- Forecasting models capable of predicting future unemployment conditions.
- Feature importance and lead-lag analyses.
- Interactive dashboards and visualizations.
- An end-to-end demonstration of Data Engineering, Analytics, Machine Learning, Visualization, Communication, and Research Design skills.

2. Stakeholders and Expected Impact

2.1 Stakeholder Groups

Policymakers and Government Agencies

Government agencies and policymakers regularly monitor labor market conditions when making fiscal and monetary policy decisions. Understanding how macroeconomic indicators relate to future unemployment may provide additional context when evaluating economic conditions and policy effectiveness.

Workforce Development Organizations

Workforce development organizations allocate training resources, workforce development programs, and employment assistance initiatives based on labor market conditions. Improved understanding of unemployment trends may support more informed resource allocation and planning decisions.

Businesses and Employers

Businesses use economic information to support hiring decisions, workforce planning, budgeting activities, and investment decisions. Understanding relationships between economic indicators and labor market conditions may improve planning during periods of economic uncertainty.

Researchers and Academic Institutions

Researchers frequently investigate labor market dynamics, economic forecasting, and macroeconomic relationships. This project contributes a reproducible analytical framework that integrates multiple economic indicators into a unified dataset suitable for future analysis.

Students and Data Science Practitioners

The project serves as a practical demonstration of how Data Engineering, Statistics, Machine Learning, Visualization, and Communication can be integrated within a single end-to-end data science project using real-world economic data.

Job Seekers and Workforce Participants

Although the project is not intended as a job market prediction tool, understanding historical relationships between economic conditions and unemployment may provide useful context regarding labor market behavior.

2.2 Shared Theme Alignment

The assigned Peer Stakeholder Product Owners share a common interest in applying data science methods to real-world decision-making problems. Their feedback is expected to influence project priorities related to model interpretability, stakeholder communication, visualization design, and the practical applicability of forecasting results. Through Studio Critiques, Iteration Reviews, and stakeholder feedback sessions, these perspectives will help ensure that project outputs remain understandable, actionable, and aligned with stakeholder needs.

2.3 Expected Impact

The purpose of this project is not to predict future economic conditions with certainty. Rather, the objective is to evaluate whether publicly available macroeconomic indicators provide meaningful information regarding future labor market conditions and whether those indicators improve unemployment forecasting performance. If successful, this project will demonstrate how publicly available economic data can be transformed into an integrated analytical asset capable of supporting economic analysis, forecasting, visualization, and communication. The project also provides a reproducible framework that may be extended to other economic forecasting problems in future work.

3. Data Sources and Access Plan

A major strength of this project is that data acquisition, integration, validation, and feature engineering have already been completed prior to proposal submission. As a result, project effort can focus primarily on analysis, forecasting, visualization, communication, and interpretation.

All datasets used in this project are publicly available, instructor-approved, and obtained through the Federal Reserve Economic Data (FRED) platform.

3.1 Source Inventory

Dataset	Source Organization	Economic Dimension	Frequency	Approximate Rows	Coverage Period	Approval Status
UNRATE	Bureau of Labor Statistics	Labor Market	Monthly	840+	1956–2025	Approved
CPIAUCSL	Bureau of Labor Statistics	Inflation	Monthly	840+	1956–2025	Approved
FEDFUNDS	Federal Reserve System	Monetary Policy	Monthly	840+	1956–2025	Approved
GDP	Bureau of Economic Analysis	Economic Growth	Quarterly	280+	1956–2025	Approved
UMCSENT	University of Michigan	Consumer Sentiment	Monthly	840+	1956–2025	Approved
USREC	National Bureau of Economic Research	Recession Indicator	Monthly	840+	1956–2025	Approved

3.2 Data Source URLs

Dataset	Source URL
UNRATE	https://fred.stlouisfed.org/series/UNRATE
CPIAUCSL	https://fred.stlouisfed.org/series/CPIAUCSL
FEDFUNDS	https://fred.stlouisfed.org/series/FEDFUNDS
GDP	https://fred.stlouisfed.org/series/GDP
UMCSENT	https://fred.stlouisfed.org/series/UMCSENT
USREC	https://fred.stlouisfed.org/series/USREC

3.3 Final Integrated Dataset

The six approved datasets were standardized, aligned by time period, validated, and integrated into a single analytical dataset.

Dataset Summary

Attribute	Value
Observation Period	April 1956 – January 2026
Total Records	838
Total Variables	40
Source Datasets	6
Duplicate Records	0
Duplicate Dates	0
Infinite Values	0
Missing Predictor Values	0
Forecast Targets	3
Engineered Variables	20+

3.4 Dataset Contribution

A significant contribution of this project is the creation of a research-ready macroeconomic dataset that integrates multiple independent economic sources into a unified analytical framework.

Unlike many educational projects that rely on a single source dataset, this project combines labor market, inflation, monetary policy, economic growth, consumer sentiment, and recession indicators into a single reproducible structure spanning nearly seventy years of United States economic history.

The resulting dataset provides a comprehensive view of economic conditions and serves as the foundation for statistical analysis, forecasting, visualization, and communication activities.

3.5 Project Readiness Statement

A major strength of this project is that a substantial portion of the Data Engineering lifecycle has already been completed prior to proposal submission.

The project enters the implementation phase with:

- Six approved datasets
- A validated analytical dataset
- 838 integrated observations
- 40 analytical variables
- More than twenty engineered features
- Three forecasting targets
- Zero missing predictor values
- Zero duplicate records
- Zero duplicate dates

As a result, project effort can focus on analysis, forecasting, visualization, communication, and interpretation rather than data acquisition and preparation.

This substantially reduces project risk and increases confidence in successful completion within the remaining capstone timeline.

4. Data Engineering Plan

4.1 Overview

Data Engineering represents a foundational component of this project. Rather than relying on a pre-built analytical dataset, the project required acquisition, validation, integration, transformation, and enhancement of multiple economic datasets originating from different organizations and publication schedules.

The objective of the Data Engineering phase was to transform raw economic indicators into a research-ready analytical dataset capable of supporting statistical analysis, machine learning, visualization, and communication.

4.2 Data Acquisition

Economic indicators were collected from publicly available Federal Reserve Economic Data (FRED) repositories and verified against their originating organizations.

Selection criteria included:

Relevance

Each dataset contributes directly to understanding unemployment and labor market behavior.

Reliability

All datasets originate from established organizations responsible for official economic reporting.

Historical Coverage

Each dataset provides sufficient historical coverage to support analysis across multiple economic cycles.

4.3 Data Standardization

Because source datasets were published at different frequencies and structures, all indicators were standardized into a common monthly analytical framework.

Standardization activities included:

- Date normalization

- Schema standardization
- Frequency alignment
- Variable naming consistency
- Temporal consistency validation

This process ensured compatibility across all economic indicators.

4.4 Data Integration

Following standardization, datasets were merged using a common monthly time dimension.

Integration activities included:

- Time-based joins
- Frequency reconciliation
- Historical alignment
- Cross-source validation

The resulting dataset contains nearly seventy years of integrated macroeconomic history.

4.5 Repository Structure, Data Management, and Version Control

4.5.1 Repository Structure

Directory	Purpose
data/	Stores all project datasets and supporting data assets.
data/raw/	Contains original source files collected from approved public data sources. Raw data remains unchanged to preserve data lineage and reproducibility.
data/processed/	Contains cleaned, standardized, integrated, and feature-engineered datasets used for analysis and modeling.
src/	Contains reusable Python scripts and data processing logic used for data ingestion, transformation, feature engineering, and analytical workflows.
notebooks/	Stores exploratory analysis, statistical analysis, feature engineering, and machine learning notebooks used throughout the project lifecycle.
deliverables/	Contains project artifacts including milestone submissions, reports, presentations, visualizations, dashboards, and final capstone deliverables.

4.5.2 Data Management Strategy

Data will be managed using a structured workflow that separates raw, intermediate, and processed data assets. Original source files will be preserved in their raw form, while all transformations and feature engineering activities will be performed on copies maintained within the processed data layer. This approach ensures transparency, reproducibility, and traceability throughout the project lifecycle.

4.5.3 Version Control and Documentation

GitHub serves as the primary version control platform for the project. All code, documentation, notebooks, and project artifacts will be maintained within the project repository and tracked through Git commits. Changes to datasets, analytical workflows, and project deliverables will be documented to support reproducibility and collaboration throughout the DS3 lifecycle.

4.6 Data Validation and Quality Assurance

Data quality validation was performed throughout the engineering process.

Validation procedures included:

Missing Value Analysis

Missing values were identified and addressed during dataset preparation.

Duplicate Detection

Duplicate observations and duplicate dates were identified and removed.

Schema Validation

Data types and structures were validated throughout integration.

Infinite Value Detection

Derived calculations were reviewed to ensure valid numerical outputs.

Final Validation Results

Validation Metric	Result
Duplicate Records	0
Duplicate Dates	0
Missing Predictor Values	0
Infinite Values	0

These results indicate that the final dataset is suitable for statistical analysis and forecasting activities.

4.7 Feature Engineering Rationale

The original economic indicators provide valuable information regarding labor market conditions; however, economic relationships are often delayed, nonlinear, and influenced by interactions between multiple variables.

To better capture these dynamics, additional analytical variables were engineered from the original datasets.

These features represent momentum effects, lagged relationships, economic stress conditions, monetary policy pressure, and future labor market outcomes.

The objective of feature engineering is to transform raw economic indicators into analytical signals that better represent economic behavior and improve forecasting performance.

4.8 Engineered Variables

Inflation Features

Variable	Description
InflationRateYoY	Year-over-year inflation growth rate.
InflationAcceleration	Measures acceleration or deceleration of inflation over time.
InflationLag3	Inflation value from three months earlier.
InflationLag6	Inflation value from six months earlier.

Labor Market Features

Variable	Description
UnemploymentChange1M	Month-over-month unemployment rate change.
UnemploymentChange3M	Three-month unemployment trend.
UnemploymentMomentum	Measures speed and direction of unemployment movement.
LaborMarketShock	Captures unusually large labor market disruptions.

Economic Growth Features

Variable	Description
GDPGrowthYoY	Year-over-year economic growth rate.
GDPGrowthMomentum	Measures acceleration or slowdown in economic growth.
GDPGrowthLag3	GDP growth value from three months earlier.
GDPGrowthLag6	GDP growth value from six months earlier.

Monetary Policy Features

Variable	Description
RealInterestRate	Inflation-adjusted interest rate environment.
MonetaryTighteningIndex	Composite measure representing restrictive monetary policy conditions.
FederalFundsLag3	Interest rate value from three months earlier.
FederalFundsLag6	Interest rate value from six months earlier.

Consumer Sentiment Features

Variable	Description
ConsumerSentimentMomentum3M	Measures changes in consumer confidence over time.
ConsumerConfidenceDeterioration	Captures periods of declining consumer confidence.

Composite Economic Indicators

Variable	Description
EconomicStressIndex	Composite indicator combining inflation pressure, unemployment momentum, and economic growth conditions to represent overall economic stress.
PolicyPressureScore	Composite measure combining inflation conditions and interest rate environments to quantify economic policy pressure.
RecessionRiskScore	Composite indicator derived from economic growth, labor market conditions, and recession signals to represent recessionary risk.

Forecast Targets

Variable	Description
FutureUnemployment_3M	Unemployment target representing labor market conditions three months into the future.
FutureUnemployment_6M	Unemployment target representing labor market conditions six months into the future.
FutureUnemployment_12M	Unemployment target representing labor market conditions twelve months into the future.

5. Planned Analysis and Methodology

5.1 Analytical Strategy

The analytical strategy is designed to answer both research questions while demonstrating competencies across Data Engineering, Statistics, Machine Learning, Visualization, and Communication.

The project will progress through four major analytical phases:

- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Forecast Modeling
- Visualization and Communication

Each phase contributes directly to answering the research questions and producing evidence-based conclusions.

5.2 Research Question Alignment

Research Question 1

To what extent can publicly available macroeconomic indicators improve forecasting of future U.S. unemployment across 3-month, 6-month, and 12-month forecasting horizons?

Planned Analysis

- Forecast target development
- Feature engineering evaluation
- Predictive model development
- Model comparison
- Forecast accuracy evaluation
- Feature importance analysis

Expected Deliverables

- Forecasting models
- Prediction performance metrics
- Model comparison results
- Forecast dashboards

Research Question 2

Which macroeconomic indicators demonstrate the strongest and most consistent historical relationships with unemployment across U.S. economic cycles between 1956 and 2025?

Planned Analysis

- Correlation analysis
- Lead-lag analysis
- Trend analysis
- Economic cycle analysis
- Feature relationship visualization

Expected Deliverables

- Correlation matrix
 - Historical relationship analysis
 - Economic cycle findings
 - Visual analytics outputs
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5.3 Exploratory Data Analysis

Prior to model development, exploratory data analysis will be performed to understand data distributions, identify anomalies, verify assumptions, and evaluate historical economic behavior.

EDA activities will include:

Dataset Profiling

- Record counts
- Variable inventory
- Data type verification
- Feature distribution analysis

Quality Verification

- Missing value assessment

- Outlier review
- Temporal consistency validation

Economic Trend Exploration

- Unemployment trends
- Inflation trends
- Interest rate behavior
- GDP growth patterns
- Consumer sentiment trends
- Recession period analysis

The objective of EDA is to ensure that downstream statistical and machine learning analyses are built upon a trusted and well-understood dataset.

5.4 Statistical Analysis

Statistical analysis will be used to evaluate historical relationships between unemployment and macroeconomic indicators.

Correlation Analysis

Measures the strength and direction of relationships between unemployment and economic indicators.

Lead-Lag Analysis

Evaluates whether changes in specific indicators consistently occur before changes in unemployment.

Trend Analysis

Examines long-term structural patterns across multiple economic cycles.

Economic Cycle Analysis

Compares indicator behavior during economic expansions and recessions.

Feature Importance Analysis

Evaluates which variables contribute most strongly to forecasting performance.

The results of these analyses will directly support Research Question 2.

5.5 Machine Learning Analysis

Machine learning models will be developed to forecast future unemployment conditions.

Forecast Horizons

- Three-Month Forecast
- Six-Month Forecast
- Twelve-Month Forecast

Candidate Models

Linear Regression

- Provides an interpretable baseline model for comparison.

Random Forest Regression

- Captures nonlinear relationships and interactions between economic indicators.

XGBoost Regression

- Provides a high-performance gradient boosting framework suitable for forecasting tasks.

- The purpose of model comparison is not simply to identify the highest-performing model but to evaluate whether macroeconomic indicators provide meaningful predictive value across different forecasting horizons.
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5.6 Model Evaluation

Forecasting performance will be evaluated using established regression metrics.

Root Mean Squared Error (RMSE)

- Measures average prediction error magnitude while placing greater emphasis on larger forecasting errors.

Mean Absolute Error (MAE)

- Measures average forecasting deviation and provides an intuitive interpretation of prediction accuracy.

R-Squared (R^2)

- Measures explanatory power and overall model fit.
 - Performance will be compared against baseline approaches using historical unemployment trends alone.
 - This evaluation framework directly supports Research Question 1.
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5.7 Visualization and Communication

The final analytical phase focuses on communicating findings through visual storytelling and stakeholder-oriented dashboards.

Planned visualizations include:

Historical Economic Trends

- Unemployment trends
- Inflation trends
- Interest rate trends
- GDP growth trends
- Consumer sentiment trends

Relationship Analysis

- Correlation heatmaps
- Lead-lag visualizations
- Economic cycle comparisons

Forecasting Analysis

- Actual versus predicted unemployment
- Forecast horizon comparisons
- Model performance dashboards
- Feature importance visualizations

The objective is to communicate technical findings to both technical and non-technical audiences.

6. Create – Observe – Analyze (DS3 Framework)

The project follows the Data-Driven Scrum (DS3) methodology through continuous Create, Observe, and Analyze cycles.

Product Backlog Item	Create	Observe	Analyze
PBI-1 Data Acquisition	Collect approved datasets	Verify completeness	Assess source suitability
PBI-2 Data Integration	Merge datasets	Observe consistency	Evaluate integration quality
PBI-3 Feature Engineering	Create analytical features	Observe distributions	Assess usefulness
PBI-4 Exploratory Analysis	Generate summaries	Observe trends	Identify relationships
PBI-5 Statistical Analysis	Produce analytical outputs	Observe indicator behavior	Interpret relationships
PBI-6 Forecast Modeling	Train models	Observe accuracy	Evaluate predictive value
PBI-7 Dashboard Development	Build visualizations	Observe stakeholder insights	Communicate findings
PBI-8 Final Reporting	Develop final artifacts	Observe feedback	Refine conclusions

The DS3 framework ensures continuous learning, structured iteration, and evidence-based decision-making throughout the capstone lifecycle.

7. Five Pillars Integration

A primary objective of this project is to demonstrate integration of all five pillars of the Master of Science in Data Science program.

7.1 Data Engineering

The project demonstrates Data Engineering through acquisition, validation, standardization, integration, and enhancement of six independent macroeconomic datasets.

Deliverables

- Integrated analytical dataset
- Data validation framework
- Feature engineering pipeline
- Dataset documentation

7.2 Analytics and Machine Learning

The project evaluates whether macroeconomic indicators improve unemployment forecasting performance through statistical analysis and predictive modeling.

Deliverables

- Baseline forecasting models

- Random Forest models
- XGBoost models
- Forecast evaluation results
- Feature importance analysis

7.3 Visualization and Communication

The project translates technical findings into stakeholder-focused visualizations and dashboards.

Deliverables

- Trend visualizations
- Economic cycle analyses
- Forecast dashboards
- Poster presentation
- Final report

7.4 Statistics and Research Design

The project applies statistical methods to evaluate historical relationships between unemployment and macroeconomic indicators.

Deliverables

- Correlation analysis
- Lead-lag analysis
- Trend analysis
- Economic cycle analysis
- Statistical interpretation

7.5 Ethics and Responsible Use

The project emphasizes transparency, reproducibility, uncertainty communication, and responsible interpretation of forecasting outputs.

Deliverables

- Ethical risk assessment
- Limitations documentation
- Reproducibility documentation
- Transparent reporting framework

8. Ethics, Risks, and Limitations

8.1 Ethical Considerations

Although this project utilizes publicly available macroeconomic data and does not involve personally identifiable information, responsible use and interpretation remain essential.

The project will adhere to the following principles:

Transparency

All data sources, transformations, feature engineering activities, and modeling decisions will be documented.

Reproducibility

Analytical workflows and project artifacts will be maintained within GitHub to support independent verification.

Responsible Communication

Forecasts and statistical findings will be presented as evidence-based estimates rather than guarantees of future economic outcomes.

Honest Interpretation

The project focuses on association and forecasting performance rather than causal inference.

8.2 Risk Assessment

Data Revision Risk

Economic indicators may be revised after publication by source organizations.

Forecasting Risk

Machine learning models may identify historical patterns that do not generalize to future economic conditions.

Economic Shock Risk

Unexpected events such as financial crises, pandemics, geopolitical conflicts, or policy interventions may create conditions not represented within historical data.

Interpretation Risk

Stakeholders may incorrectly interpret relationships as evidence of causality.

These risks will be addressed through transparent documentation, model validation, reproducible analytical workflows, and careful interpretation of forecasting results.

8.3 Project Limitations

Observational Data

The dataset is observational and therefore cannot establish causal relationships.

National-Level Focus

The analysis focuses on national economic indicators and may not reflect regional labor market conditions.

Historical Dependence

Forecasting models rely on historical relationships that may evolve over time.

Public Data Constraints

The project is limited to publicly available economic indicators included within the approved dataset.

9. Timeline and Milestones

Time Period	Activities	Deliverables
Proposal Phase	Literature review, research design, data engineering	M1 Proposal
Weeks 5–6	Exploratory analysis and validation	Preliminary findings
Week 7	Dataset summary and analytical review	M2 Data Summary

Weeks 8–9	Statistical analysis and forecasting development	Analytical outputs
Week 10	Visualization development and model refinement	M3 Poster Draft
Weeks 11–12	Dashboard development and communication artifacts	Visual analytics deliverables
Week 12	Report drafting	M4 Rough Draft
Weeks 13–14	Final refinement and stakeholder feedback	Final deliverables
Week 14	Poster presentation and final submission	M5 Final Submission

10. DS3 Process and Project Tooling

10.1 Project Resources

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Owner Product Lead: Manish R. Kallu

Peer Stakeholder Product Owners:

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Studio Session: 2

Studio Formation Date: May 25, 2026

10.2 Individual Contribution Statement

- This is an individual capstone project.
- The literature review, research question development, dataset selection, data acquisition, data validation, feature engineering, analytical design, forecasting methodology, dashboard development, and proposal preparation were completed by the project owner.
- Peer Stakeholder Product Owners participate exclusively in the DS3 feedback process and do not contribute to the research, engineering, or analytical components of the project.

10.3 Iteration and Feedback Process

- Feedback received through Studio briefs from Peer Stakeholder Product Owners, and Instructor Reviews will be incorporated into future project iterations.
- This process supports continuous improvement of analytical methods, communication strategies, visualizations, and final deliverables.

11. Conclusion

This project investigates whether publicly available macroeconomic indicators improve understanding and forecasting of future U.S. unemployment conditions.

Beginning as a study of historical unemployment trends, the project evolved through literature review, dataset evaluation, and feasibility analysis into a broader investigation of labor market forecasting and economic relationships. The resulting analytical framework integrates six approved economic datasets spanning nearly seventy years of United States economic history and combines Data Engineering, Statistics, Machine Learning, Visualization, and Communication into a single reproducible workflow.

The project contributes a validated analytical dataset containing 838 observations and 40 variables, engineered economic indicators, forecasting models, statistical insights, and stakeholder-oriented visualizations.

By leveraging publicly available economic data and transparent analytical methods, the project seeks to demonstrate how modern data science techniques can be applied to understanding complex economic systems.

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